



Department of Electrical and Electronics Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: V Semester B.E EEE

Course Code & Title: EE3591 – Power Electronics

Name of the Faculty member : S. Meenakshi Sundaravel

Innovative Practice Description

❖ **Unit / Topic:** Units – I to V/ Symbol and Applications of various power electronic devices

- **Course Outcome:** CO1, CO2, CO3, CO4, CO5

- **Topic Learning Outcome:** 1a, 2a, 3a, 4a, 5a

- **Activity Chosen:** One minute paper

- **Justification:**

Students have to familiarize with the name, abbreviation, symbols and applications of various power electronic devices. Since, one minute paper activity provides a conceptual bridge between successive class periods; it helps the learners to summarize and have a quick revision of all the 5 units in this course. Also, it improves the quality of class discussion by having students writing briefly about the entire course.

Time Allotted for the Activity: 05 minutes

- **Details of the Implementation:**

At the end of the class, students were asked to write about the symbol and applications of various power electronic devices discussed in all the 5 units. The students expressed the understood content and the content which were not clear in that particular topic. This activity shows whether the students can able to understand the specific topic and their involvement in the particular class.

- **CO – PO / PSO mapping:**

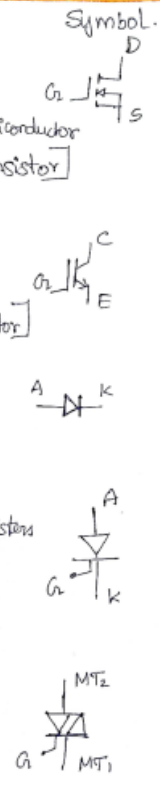
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	-	2	-	-	-	1	1	1	1	2	-	1
CO2	3	2	1	-	2	-	-	-	1	1	1	1	2	-	1
CO3	3	2	1	-	2	-	-	-	1	1	1	1	2	-	1
CO4	3	2	1	-	2	-	-	-	1	1	1	1	2	-	1
CO5	3	2	1	-	2	-	-	-	1	1	1	1	2	-	1

- **PO / PSO mapped:**

(1 – Low 2 – Moderate 3 – High)

Innovative practice	PO1
	3
Justification for correlation	The students can have the strong fundamental knowledge.

• **Images / Screenshot of the practice:**

Innovative Teaching Method Execution		
Symbol and Applications of various power electronic devices – One minute paper		
Devices 1) MOSFET [Metal Oxide Semiconductor Field Effect Transistor] 2) IGBT [Insulated Gate Bipolar Transistor] 3) Power Diode 4) SCR (or) Thyristor [Silicon Controlled Rectifier] 5) TRIAC [Triode AC]	Symbol 	Application * SMPS * Chopper * Inverter * Rectifiers * Clippers * Controlled Rectifiers * Voltage controllers * Voltage controllers * Induction heating

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- ✓ Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

❖ **Benefit of the practice:**

- ✓ Students can able to attend the question even in the questions are in indirect form.
- ✓ Students can able to explain the concepts in examination without any confusion.

❖ **Challenges faced in implementation:**

- ✓ Time utilization for conducting activity.

References:

1. Rashid M.H., Power Electronics Circuits, Devices and Applications, Prentice Hall India, 3rd Edition, New Delhi, 2004.
2. P.S. Bimbhra, Power Electronics, Khanna Publishers, Third Edition 2003.

Signature of Faculty Member

HOD



Department of Electrical and Electronics Engineering Academic Year 2024 – 2025 (Odd Semester)

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Name of the Faculty member: Mr. S. Meenakshi Sundaravel

Innovative Practice Description

- **Unit / Topic:** Unit III / Capacitor filter for low power rectifiers.
- **Course Outcome:** CO 3
- **Unit Outcome:** 3d
- **Activity Chosen:** Virtual Lab
- **Justification:**

Virtual lab enables a virtual teaching and learning environment which develops students' practical knowledge. It is one of the most important eLearning tools that allow the student to perform various experiments without any constraints to place or time. Through this VLAB, the student can understand the operation and characteristics of capacitor filter used in rectifiers.

Time Allotted for the Activity: 15 minutes

- **Details of the Implementation:**

The students were asked to do the following steps,

1. Take 4 diodes, Load resistor of 1 KOhms and capacitor of 102 μ F.
2. Connect to AC Voltage Source of 50 Hz, 12 V.
3. Click on 'ON' button to make the circuit on.
4. Click on 'Sine Wave' button to observe the input waveform.
5. Click on 'Run Simulation' button to observe the filtered waveform.
6. Vary the amplitude using the controllers.
7. Observe the corresponding waveform.
8. Channel 1 shows the input wave, Channel 2 shows the output wave and Dual shows both the input and output wave.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO3	3	2	1	-	2	-	-	-	1	1	1	1	2	-	1

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:

Innovative practice	PO5
	2
Justification for correlation	The students can effectively use the virtual lab and acquire knowledge about the working of C-filters.

- Images / Screenshot of the practice:

Innovative Teaching Method Execution

Operation of Full wave rectifier with Capacitor filter – Virtual Lab



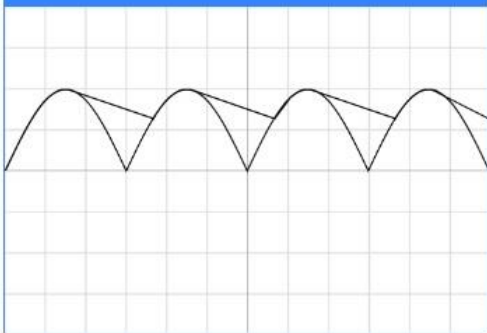
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Capacitive Rectification for Full Wave Rectifier

INSTRUCTION

GRAPH PLOT



Channel 1
Channel 2
Ground
Dual

CALCULATION

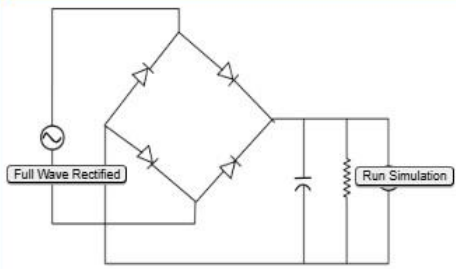
Measure the V_m

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$V_{dc} = \frac{V_m}{\pi}$$

$$\text{Ripple Factor} = \frac{V_r}{V_{dc}} \quad \text{Since, } V_{ac} = \frac{\sqrt{(V_1^2 - V_2^2)}}{V_{dc}}$$

CIRCUIT



Off

CONTROLS

V_{Pch1} : 1 V

Position Y-Axis: 0

Phase: 0 Deg

Frequency: 1000 Hz

V_{Pch2} : 1 V

Position Y-Axis: 0

Phase: 0 Deg

Frequency: 1000

- Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- ✓ More questions were asked to the students and came to know that they understood the concept well and the usage of virtual lab.
- ✓ They could be able to plot the filtered output waveform.

❖ ***Benefit of the practice:***

Beyond knowing the theoretical concepts, it is important to implement the concepts practically. So, knowing simulation is also needed to improve the technical knowledge of the students. Students can correlate the concepts studied theoretically with the experiments they simulated.

❖ ***Challenges faced in implementation:***

I planned the activity for 15 minutes. But it takes 30 minutes to implement.

References:

1. Rashid M.H., Power Electronics Circuits, Devices and Applications, Prentice Hall India, 3rd Edition, New Delhi, 2004.
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Innovative Practice Description

- **Unit / Topic:** Unit IV / Controlled Rectifiers
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 4c.
- **Activity Chosen:** Active Learning Method – MATLAB Simulation

- **Justification:**

Controlled rectifiers uses thyristor or SCR for AC to DC conversion. These can be classified as single phase rectifier and three phase rectifier. Single phase rectifiers are classified as 1- Φ half controlled rectifier or 1- Φ semiconverter and 1- Φ fully controlled rectifier. Three phase rectifiers are classified as 3- Φ half controlled rectifier or 3- Φ semiconverter and 3- Φ fully controlled rectifier. After teaching 1- Φ fully controlled rectifier and 1- Φ semiconverter, I had conducted a simulation activity for making the students to understand the operation of both the types in practical mode with various loads and also enhance their learning level and as a teacher I can judge the understanding level of the students.

- **Time Allotted for the Activity:** 40 minutes

- **Details of the Implementation:**

Total Strength is 63.

At the end the Class (Last 10 minutes)

- ✓ I asked the students to simulate the 1- Φ fully controlled rectifier and 1- Φ semiconverter for various loads for 30 minutes.
- ✓ Then I asked them to obtain the controlled output voltage waveform of them for another 5 minutes.
- ✓ Finally, I randomly asked students about the operation of 1- Φ fully controlled rectifier and 1- Φ semiconverter for 5 minutes.

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 5	PO10	PO12	PSO1	PSO3
CO 4	3	2	1	2	1	1	2	1

(1 – Low 2 – Moderate 3 – High)



- PO / PSO mapped:

Innovative practice	PO 5
	2
Justification for correlation	Students have utilized the modern tool MATLAB for performing simulation. Hence, this activity is correlated with level 2.

- Images / Screenshot of the practice:



Reflective Critique:

- ❖ *Feedback of practice from students and other stakeholders:*

Students were able to perform the simulation of 1- Φ fully controlled rectifier and 1- Φ semiconverter and also they said that they understood the functions of them.

- ❖ *Benefit of the practice:*

1. Students can able to understand the impact of engineering solution on society.
2. Students understood the concept which was reflected from their answers for the questions I have asked.
3. The success of the activity was evaluated by asking the same question during IAT – I – **Around 80% of students answered.**
4. Students can able to explain the concepts in detail in the examination.

- ❖ *Challenges faced in implementation:*

1. Time utilization for conducting activity, slow learners were not able to perform the simulation quickly.

References:

- ✓ P.S.Bimbira “Power Electronics” Khanna Publishers, Third Edition, 2003.

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Innovative Practice Description

- **Unit / Topic:** Unit V / AC Phase Controllers
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 5c.
- **Activity Chosen:** Active Learning Method – Demonstration using Modern tool (FLUKE Power Quality Analyzer)
- **Justification:**

AC voltage controllers (ac line voltage controllers) are employed to vary the RMS value of the alternating voltage applied to a load circuit. An AC voltage controller is a type of thyristor power converter which is used to convert a fixed voltage, fixed frequency ac input supply to obtain a variable voltage ac output. The RMS value of the ac output voltage and the ac power flow to the load is controlled by varying (adjusting) the trigger angle ' α '. The ac voltage controllers are classified into two types based on the type of input ac supply applied to the circuit, single phase AC controllers and three phase AC controllers. Single phase ac controllers operate with single phase ac supply voltage of 230 V RMS at 50 Hz in our country. Three phase ac controllers operate with 3 phase ac supply of 400 V RMS at 50Hz supply frequency. Applications of AC Voltage Controllers are Lighting / Illumination control in ac power circuits, Induction heating, Industrial heating & Domestic heating, Transformers tap changing (on load transformer tap changing) and Speed control of induction motors (single phase and poly phase ac induction motor control).

- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**

Total Strength is 63.

At the end the Class (Last 10 minutes)

- ✓ I have demonstrated the students about the operation of 1- Φ AC Voltage controllers using FLUKE Power Quality Analyzer for about 30 minutes.



- ✓ Then I have demonstrated to observe the output voltage and current waveforms, how to get FFT analysis waveform of output voltage include THD value and all harmonic component values with the help of FLUKE Power Quality Analyzer to them for another 5 minutes.
- ✓ Finally, I randomly asked students about the operation of 1- Φ full wave AC voltage controller for 5 minutes.

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 5	PO10	PO12	PSO1	PSO3
CO 4	3	2	1	2	1	1	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 5
	2
Justification for correlation	Students have utilized the modern tool FLUKE Power Quality Analyzer for analyzing the performance of a power electronic controller. Hence, this activity is correlated with level 2.

- **Images / Screenshot of the practice:**





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Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students were able to differentiate the usage of CRO/DSO with FLUKE Power Quality Analyzer and realized the importance of FLUKE meter in various aspects.

❖ *Benefit of the practice:*

1. Students can able to understand the impact of engineering solution on society.
2. Students understood the concept which was reflected from their answers for the questions I have asked.
3. The success of the activity was evaluated by asking the same question during IAT – II – **Around 80% of students answered.**
4. Students can able to explain the concepts in detail in the examination.

❖ *Challenges faced in implementation:*

1. Time utilization for conducting activity.

References:

- ✓ P.S.Bimbra “Power Electronics” Khanna Publishers, Third Edition, 2003.

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